

Week 1 – Report

Intern: Achref Elarbi | **Role:** Capstone Data Scientist Intern | **Company:** Realwise (Real Estate)

Day 1 (2/10/2026)

Goal of the Day:

My main goal was to discover the real estate industry, understand how the market functions, and identify the key challenges faced by different stakeholders like buyers, sellers, investors, agents, and property managers.

1 Overview of Real Estate

Through my research, I learned that real estate is essentially **land and anything permanently attached to it**, including homes, offices, factories, and specialized buildings. I discovered the main types of real estate:

- **Residential:** Homes, condos, townhouses, villas
- **Commercial:** Shops, offices, car parks
- **Industrial:** Factories, warehouses, distribution centers
- **Special Purpose:** Schools, museums, railways
- **Land:** Farms, open land, undeveloped areas

I also noted the key roles in the industry:

- **Agents/Realtors:** They assist buyers, sellers, and investors with pricing, paperwork, and finalizing deals.
 - **Property Managers:** They handle rental properties, maintenance, and tenant relationships.
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2 How the Real Estate Market Works

I learned that the market operates on the balance of **supply and demand**. Properties are bought, sold, or rented based on how many are available and how many people want them. The process involves listings,

Top 7 Problems & Solutions

1. **High lead costs / low conversion** → Use CRM to track and follow up leads.
2. **No clear sales process** → Set up a step-by-step workflow for deals.
3. **Poor client communication** → Keep clients updated and respond promptly.
4. **Not enough listings** → List more properties and create eye-catching ads.
5. **Handling rejections** → Stay patient, follow up, and maintain relationships.
6. **Missed technology opportunities** → Use social media, online listings, virtual tours, and CRM software.
7. **Unrealistic pricing (sellers)** → Educate sellers on market trends and set fair prices.

Takeaways / Personal Reflection:

- Real estate is **complex**, with multiple players and challenges.
- **Technology and data** are essential to improve lead conversion, pricing, and market insights.
- Understanding customer problems is key for **data-driven solutions**, which I can apply in my capstone project.

Conclusion:

Day 1 gave me a solid foundation of the real estate market, customer pain points, and operational challenges. I now see where data science can help improve decision-making and efficiency.

Day 2 (2/11/2026)

Goal:

Understand PropTech and where data science fits in specifically predictive analytics, automation, and investment decisions.

1 What Is PropTech?

PropTech is basically any tech that makes real estate better. It touches three main phases:

- **Pre-Development:** Apps that help developers, investors, and planners communicate and share info easily.
 - **Construction:** AI, drones, and robots help with cost estimates, scheduling, and tracking progress. Adoption is slow because accountability is tough.
 - **Property Management:** Software, IoT sensors, and dashboards for collecting rent, tracking maintenance, talking to tenants, and automating marketing.
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2 How It Changes Real Estate

PropTech impacts three big areas:

- **Property Search & Investment:** Virtual tours, detailed listings, and AI recommendations save time and help investors make smarter decisions. Blockchain makes transactions secure and transparent.
 - **Property Management:** Less paperwork, more automation. Tools handle tenant screening, rent, maintenance requests, energy tracking, and security. Smart contracts cut down fraud.
 - **Data & Insights:** Machine learning shows market trends, property values, and portfolio risks—helping investors get better returns.
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3 PwC PropTech Report (Mid-2025)

Key takeaways from the report:

- **AI everywhere:** Predictive maintenance, valuations, leasing, marketing. AI-focused startups get the most funding.
 - **Hot categories:** Construction, Property Management, Investment & Financing.
 - **What's growing:** M&A deals are back, construction tech is booming, full-stack platforms (finance + leasing + maintenance), and IoT needs real-time data.
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Takeaways:

- PropTech is making real estate faster, clearer, and more data-driven.
- AI, IoT, and analytics are where data science can shine.
- The key is understanding problems, automating workflows, and building predictive models.

✓ Bottom line:

Day 2 showed me how tech is reshaping real estate and where I can apply data science to improve operations and investment decisions.

Day 3 (2/12/2026)

Goal of the Day:

My main goal was to explore global and European PropTech solutions, understand what problems they solve in real estate, and identify where data science and AI could add real value.

1 Overview of PropTech Solutions

Today I mapped out the main categories of PropTech and looked at real examples to see how they address industry challenges:

- **Property Search & Marketplaces:** Digital platforms like Zillow-type systems connecting buyers, renters, agents, and property owners.

- **Property Management Software:** Tools for landlords and facility managers to track maintenance and operations.
- **Smart Buildings & IoT:** Sensors and automation for building access, energy use, and operational efficiency.
- **FinTech for Real Estate:** Platforms for investments, mortgages, and portfolio management.
- **Construction Tech:** AI-powered site monitoring and modular building solutions.
- **Rental & Leasing Automation:** AI chatbots and dashboards to manage tenants and short-term rentals.
- **Data & Analytics Platforms:** Market intelligence, predictive pricing, and decision-making tools.

Example platforms:

- **ButterflyMX:** Smart building access system
- **Infraspeak (Portugal):** Facility management with IoT
- **EliseAI:** AI leasing chatbot automating 80% of inquiries
- **Guesty / Hostaway:** Unified short-term rental management
- **Agora:** Investment and portfolio analytics
- **OpenSpace:** AI digital twins for construction monitoring
- **CorkBrick (Portugal):** Sustainable modular construction

2 Mapping Problems to Solutions

I connected each platform to the real problem it solves:

Company	Problem	Solution
ButterflyMX	Old, fragmented intercoms	Cloud-based smart access system
Infraspeak	Manual maintenance tracking	IoT sensors + digital platform
EliseAI	Leasing staff overloaded	AI chatbot automates inquiries

Company	Problem	Solution
Guesty	Complex multi-platform management	Unified dashboard for rentals
OpenSpace	Hard to track construction progress	AI-generated digital twins

Data science perspective: Many solutions already use automation, pattern recognition, or predictive elements. This shows where ML/AI can optimize processes, forecast trends, or create insights.

3 Strengths and Limitations

Strengths:

- Automation saves time and reduces errors
- Scalable solutions with proven traction
- Some platforms provide data-driven insights
- Integration improves operational efficiency
- Cost savings for property managers and investors

Limitations:

- High upfront costs for hardware or software
- Adoption barriers in conservative real estate markets
- Data fragmentation and inconsistent quality
- Regulatory complexity in Europe
- Analytics often limited to dashboards rather than predictive insights

Examples:

- **EliseAI:** Cuts workload by 80%, but small agencies may not benefit much
- **ButterflyMX:** Modern access and scalable, but requires hardware
- **OpenSpace:** Tracks progress automatically, but depends on consistent photo capture

4 European & Portuguese Context

Europe has ~3,500 PropTech startups, with Portugal showing strong innovation in:

- Facility and maintenance management (Infraspeak)
- Sustainable construction tech (CorkBrick)
- Digital tools for real estate management (Relive)

Challenges include fragmented markets, regulatory differences, slow adoption, and limited public data.

5 Opportunities as a Data Scientist

- **Underutilized Data:** Many platforms collect data but only provide basic dashboards
- **AI/ML Potential:** Predictive pricing, tenant risk scoring, maintenance forecasting, dynamic investment analysis
- **Data Gaps:** Market and behavioral data is incomplete, fragmented, or inconsistent
- **Integration Issues:** Systems often don't communicate, creating siloed data

Key takeaway: My contribution isn't just another dashboard. It's about building predictive models, creating pipelines, uncovering patterns, and turning data into actionable insights. Europe and Portugal especially need better integrated, analytics-driven solutions.

✓ Conclusion:

Day 3 gave me a clear view of the PropTech landscape, real problems being solved, and where data science can make a real difference. Automation exists, but predictive and integrated analytics are still underdeveloped—this is where I can add value.

Day 4 (2/13/2026)

Goal of the Day

Analyze the Portuguese and European PropTech market, compare it with global leaders (US & UK), and identify local gaps and opportunities for data science.

1 Portugal vs Global PropTech

Globally (US/UK), PropTech focuses on:

- Predictive pricing
- Investment optimization
- Algorithmic buying (iBuying)
- Large-scale AI experimentation
- Strong venture capital backing

Portugal follows a different path.

The ecosystem is growing but focuses more on:

- Facility management
- Predictive maintenance
- Smart buildings & IoT
- Sustainability & ESG compliance
- Operational efficiency

Portugal is not behind — it is more compliance-driven and sustainability-focused.

Innovation is shaped by:

- GDPR
- EU AI Act
- Conservative real estate culture

This results in slower but more regulated development.

2 AI Adoption in Portugal

AI is actively used in:

- Predictive maintenance
- Energy optimization
- Smart building monitoring
- Operational automation

Strength areas:

- IoT integration
- Maintenance forecasting
- Sustainability analytics

Weaker areas:

- Hyper-local predictive pricing
- Construction lifecycle AI
- Advanced tenant risk scoring
- Investment simulation tools

AI exists, but it is mainly operational rather than predictive or financial.

Key Market Gaps

- **Predictive pricing deficit:** Platforms are listing-focused, not insight-driven.
- **Data fragmentation:** CRM, maintenance, financial, and IoT systems are siloed.
- **Underdeveloped construction AI:** Limited cost forecasting and digital twin adoption.
- **Regulatory limits:** GDPR and EU AI Act restrict behavioral risk models.

Main issue: lack of integrated data pipelines.

Opportunities for Data Science

Strong potential in:

- Predictive valuation models
- Maintenance forecasting (time-series modeling)
- Energy optimization analytics
- Unified data architecture
- GDPR-compliant AI systems

The biggest opportunity is transforming fragmented operational data into predictive decision-support systems.

Conclusion

Portugal's PropTech market is growing and innovation is steady, especially in sustainability and facility management.

However, predictive analytics and system integration remain underdeveloped.

As a data scientist, the opportunity lies in building intelligent, integrated, and compliant AI solutions that move the market from dashboards to predictive insights.